

CUSTOMER JOURNEY MAP

Date	04 July 2026
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Team id	

The Customer Journey Map illustrates the end-to-end process of analysing electricity consumption patterns in India using Tableau. It outlines each stage, from identifying the problem and collecting data to cleaning, analysing, building interactive dashboards, and generating insights. The journey highlights key challenges, user actions, observations, and opportunities for improving energy planning and decision-making through data visualization. This map helps understand how raw electricity consumption data is transformed into meaningful insights that support efficient energy management and policy decisions.

Key Journey Steps

1. Identify the problem statement.
2. Collect electricity consumption data.
3. Clean and validate the dataset.
4. Analyse state-wise and region-wise consumption.
5. Compare 2019 and 2020 electricity usage.
6. Evaluate lockdown vs. recovery consumption.
7. Create interactive Tableau dashboards.
8. Generate insights and recommendations.

ELECTRICITY CONSUMPTION ANALYSIS OF INDIAN STATES (2019-2020)

Understanding electricity consumption patterns and the impact of COVID-19 lockdown using Tableau

SCENARIO		A data analyst / researcher wants to understand state-wise electricity consumption in India during 2019 - 2020 and understand the impact of COVID-19 lockdown using dashboards and visualizations.						
STEPS	Understand Objective	Collect Data	Prepare & Clean Data	Explore Data	Create Visualizations	Build Dashboard	Analyze & Derive Insights	Share & Present Findings
 Goals & Motivations	- Understand the problem statement - Identify key questions to answer - Define scope of analysis	- Gather electricity consumption data for 2019-2020 - Ensure data is reliable and complete	- Handle missing values - Remove duplicates - Standardize data for analysis	- Explore data to understand patterns - Identify key trends and variations	- Choose the right charts and maps - Visualize trends and comparisons effectively	- Combine visuals into interactive dashboard - Make it user-friendly	- Interpret the visualizations - Derive meaningful insights - Understand impact of lockdown	- Present findings to stakeholders - Support decision making with data - Suggest actions
 Interactions (Things / Places / People)	- Project Guide / Mentor - Research Papers - Problem Statement - Past Reports	- Data Sources (MoP, Govt. reports) - CSV / Excel Files - Kaggle / Data Portals	- Excel / SQL - Data Cleaning Tools - Data Validation Techniques	- Tableau - Summary Tables - Basic Charts - Data Analyst	- Tableau - Maps, Bar Charts, Line Charts, Heat Maps, Pie Charts	- Tableau Dashboard - Filters & Parameters - Interactive Features	- Analyst - Domain Knowledge - Statistical Understanding	- Stakeholders - Presentations - Reports (PDF/PPT) - Tableau Public (optional)
 Positive Moments	Clear understanding of what needs to be analyzed.	Finding relevant and complete data from trusted sources.	Clean and structured data ready for analysis.	Discovering interesting trends and variations in the data.	Visuals clearly show patterns and make data easy to understand.	Dashboard looks interactive and insightful.	Deriving valuable insights that explain the impact of COVID-19 lockdown.	Stakeholders appreciate the findings and insights.
 Negative Moments	Unclear requirements or scope can cause confusion.	Data may be incomplete, missing or inconsistent.	Time-consuming data cleaning and handling missing values.	Large dataset can be difficult to explore.	Choosing the wrong chart may lead to misinterpretation.	Dashboard may become cluttered or difficult to navigate.	Misinterpretation of data can lead to wrong conclusions.	Technical insights may not be easy for non-technical stakeholders.
 Areas of Opportunity	Define clear objectives and key questions at the start.	Use reliable government sources and automate data collection if possible.	Use efficient data cleaning methods and tools to save time.	Use pivot tables and summary statistics for effective exploration.	Follow best practices for data visualization and storytelling.	Keep dashboard simple, interactive and focused on key metrics.	Validate insights with domain knowledge and additional data.	Use clear visualizations and simple language for better communication.
 PROJECT OUTCOME: An interactive Tableau dashboard that helps understand electricity consumption patterns across Indian states and the impact of COVID-19 lockdown.								